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RESEARCH ARTICLE

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A population-based legacy study of myasthenia gravis in Iceland: insights from a small Arctic nation

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ABSTRACT

Myasthenia gravis (MG) is a rare autoimmune neuromuscular disorder for which population-based estimates remain scarce, particularly in small, homogeneous settings. Leveraging Iceland's centralized healthcare, we reassessed MG epidemiology and clinical features in a nationwide legacy cohort. Cases diagnosed from 1981 to 2002 were identified through hospital databases, private neurologists, immunology laboratory records and the national MG association. Inclusion required characteristic symptoms plus at least one confirmatory test (response to acetylcholinesterase inhibitors, abnormal electrophysiology, or positive antibodies). Point prevalence was calculated on 31 December 2002, and incidence for 1997–2002. In 2003, patients were evaluated to document diagnostic delay, treatment patterns and disease severity; descriptive statistics and non-parametric tests were applied. Twenty-six patients met criteria, yielding a prevalence of 9.0 per 100,000 and a mean annual incidence of 0.78 per 100,000. The female-to-male ratio was 1.2:1; mean age at onset was 43 years overall (29 in women, 59 in men). Ocular symptoms predominated at presentation. Nineteen percent experienced diagnostic delays exceeding three years. By 2003, 54% had generalized MG and 27% were in complete remission off medication. This historical cohort aligns with global trends of rising prevalence and illustrates the value of legacy datasets for rare disease surveillance.

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Myasthenia gravis; prevalence; incidence; rare diseases; population-based study; clinical characteristics; Iceland

Introduction

Myasthenia gravis (MG) is an autoimmune neuromuscular disorder, classified among rare diseases by the European Medicines Agency (EMA) as affecting fewer than 1 in 2000 individuals [1]. Like many rare conditions, MG is understudied due to low case numbers, geographic variability and the inherent difficulties in gathering robust epidemiological data. These limitations hinder the development of a comprehensive understanding of the disease's epidemiology, natural history and treatment responsiveness [2]. In this context, every available data set, whether newly collected or historical, holds significant value. Increasing attention is being awarded to underutilized sources such as legacy cohorts, which allow temporal comparisons, serve as benchmarks for trends in incidence and prevalence, and offer critical insight into evolving diagnostic practices and disease trajectories over time [3,4].

MG is a well-recognized and clinically complex neuromuscular disorder, characterized by fluctuating weakness in voluntary muscles. Symptoms typically worsen with physical exertion and improve with rest, often presenting as minimal in the morning and progressively intensifying throughout the day [5]. While the exact cause remains unknown, it is well-established that a sizable proportion of patients produce autoantibodies targeting acetylcholine receptors, leading to impaired neuromuscular transmission [6]. These immunological changes are closely linked to thymic pathology, including thymomas in approximately 10–12% of patients [7]

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and follicular hyperplasia in up to 70% [8]. These foundational findings continue to shape diagnostic criteria and guide therapeutic approaches.

The most common initial symptoms involve ocular muscles resulting in ptosis and/or diplopia, but the disease may also affect bulbar, limb and trunk muscles. In severe cases, respiratory compromise may result from involvement of the diaphragm and intercostal musculature [9]. MG can present at any age but is rare before age 15 or after 70 [10]. Women are more commonly affected between the ages of 20–30, while men are typically diagnosed later, between ages 50–70, with an overall male-to-female ratio of approximately 1:2 [11]. Treatment is based on three primary pillars: acetylcholinesterase inhibitors, immunosuppressive therapy and thymectomy [9].

Over the last two decades, the prevalence and incidence of MG have continued to rise globally, driven by improved recognition, diagnostic tools and demographic shifts such as population aging [10,12,13]. To shed light on these historical data, baselines are needed. Iceland's centralized healthcare system and comprehensive medical record-keeping provide strength and reliability to legacy cohort studies. Importantly, analyzing data from earlier diagnostic sheds light on how diagnostic delays, treatment access and remission patterns have evolved, especially prior to the widespread use of newer antibody tests (e.g. anti-MuSK, anti-LRP4). Such insights are particularly valuable in rare disease contexts, where national registries and longitudinal cohorts are pivotal for disease surveillance, resource allocation and shaping policy [14,15].

Over recent decades global epidemiological studies have demonstrated wide variations in MG prevalence and incidence across countries, attributed to greater awareness, improved diagnostics and population aging [16]. Iceland, despite its small population, has contributed meaningfully to this field. The first population-based MG study, conducted in Iceland [17] was a landmark effort at a time when systematic data on MG were limited. This early investigation reported a prevalence of 6.4 per 100,000 and an incidence of 0.41 per 100,000. A follow-up, unpublished study in 1991 showed a slight increase in prevalence to 6.8 per 100,000, suggesting epidemiological stability across nearly three decades [18]. The present study aimed to build upon this foundational work by providing a newer timepoint for the assessment of MG prevalence and incidence in Iceland, alongside an in-depth characterization of clinical presentation, diagnostic practices, disease course and treatment patterns. The data, originally collected two decades ago, are published here for the first time. Although not a longitudinal follow-up of previous cohorts, this analysis serves as a legacy dataset, capturing a distinct historical period in MG care and diagnosis in Iceland, and thereby enables valuable comparison with both earlier and future studies.

Methods

Design

This historical data aimed to establish the point prevalence and incidence of MG in Iceland during the period (1997–2002), as well as to describe the clinical characteristics of the affected population. To ensure comprehensive and transparent reporting, this study adhered to the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) checklist [19].

Setting and participants

Patients with MG in Iceland were identified through electronic records from the Neurology Department of Landspítali University Hospital using diagnostic codes ICD-9 358.0 and ICD-10 G70.0. The data search covered the period from January 1, 1981, to December 31, 2002. Letters were sent to all practicing neurologists in Iceland and to the chief physicians of all primary healthcare centres across the country. Records from the Immunology Laboratory at Landspítali, the only such laboratory in Iceland, were reviewed for positive antibody test results from 1984 to 2002. The Icelandic MG patient association was informed about the study, and members encouraged to contact the research team.

After receiving permission from each patient's treating physician, an information letter and consent form were sent to eligible patients. Subsequently, patients were invited to attend a clinical interview and standardized examination at the Landspítali outpatient clinic. Those who were unable to attend provided information about their medical history via telephone, and the most recent clinical assessment from their

physician was reviewed. This evaluation of the study cohort was conducted in February 2003. Population data were obtained from Statistics Iceland.

Inclusion in the study cohort required that the diagnosis be supported by at least one of the three diagnostic criteria mentioned earlier: a positive response to acetylcholinesterase inhibitors, a decremental muscle response to repetitive nerve stimulation, or a positive antibody test, in addition to characteristic clinical symptoms. Congenital and drug-induced myasthenic syndromes were excluded.

For clinical classification, the Osserman staging system was used [20]:

- A = Asymptomatic without medication
- I = Ocular symptoms only (diplopia/ptosis)
- IIA = Mild generalized symptoms with minimal impact on daily life
- IIB = Generalized symptoms causing significant functional impairment
- III = Rapidly progressive generalized symptoms with prominent bulbar involvement
- IV = Severe generalized symptoms with respiratory failure (myasthenic crisis)

For comparability with contemporary epidemiological studies, Osserman classifications were retrospectively mapped to the Myasthenia Gravis Foundation of America (MGFA) clinical classification [21] based on documented symptom distribution and severity at assessment.

Data analysis

Descriptive statistics were used to summarize the demographic and clinical characteristics of the cohort. Prevalence was calculated as point prevalence on 31 December 2002, when the population of Iceland was 288,471, and expressed per 100,000 inhabitants based on national census data. Incidence was calculated for the five-year period from 31 December 1997, to 31 December 2002, and reported per 100,000 inhabitants per year. Age at diagnosis and symptom onset were described using medians, interquartile ranges (IQR) and full ranges. To explore associations between categorical variables (e.g. MG subtype and gender, or MG subtype and treatment pattern), Fisher's exact test was used. Associations between continuous variables and MG subtype (e.g. age at diagnosis) were assessed using the Mann–Whitney U test. The distribution of age at diagnosis by MG type was visualized using a Seaborn violin plot, which integrates a kernel density estimate with a boxplot to display both the overall data distribution and key summary statistics, including the median and interquartile range. Statistical analyzes were conducted using SPSS, a p-value < 0.05 was considered statistically significant.

Ethical considerations

The National Bioethics Committee (VSN-a2002110045) and the Icelandic Data Protection Authority (S1115/2003) approved the study.

Results

Study population

Responses were received from all neurologists in Iceland and from 49 of 59 (83%) general practitioners before the study period ended. Including the data from Landspítali University Hospital, a total of 57 patients were initially identified. After further evaluation, 31 individuals were excluded, resulting in a final cohort of 26 patients. Of these, 21 patients attended clinical examinations and interviews.

Prevalence

The point prevalence of MG in Iceland as of 31 December 2002, was 26 out of 288,471 inhabitants alive that day, equating to 9.0 per 100,000. The sex-specific prevalence was 9.7 per 100,000 for women and 8.3 for men, with a male-to-female ratio of 1:1.2 (see Table 1).

Table 1. Age- and Sex-Specific Prevalence of MG per 100,000 in Iceland.

Men			Women			Total			
Age	n	Population	Prevalence*	n	Population	Prevalence*	n	Population	Prevalence*
0–14	0	33,768	0	2	32,371	6.2	2	66,139	3
15–24	0	21,677	0	0	21,050	0	0	42,727	0
25–34	0	20,872	0	1	20,338	4.9	1	41,210	2.4
35–44	1	21,428	4.7	3	21,447	14	4	42,875	9.3
45–54	1	18,938	5.3	2	18,281	10.9	3	37,219	8.1
55–64	2	12,380	16.2	4	12,130	33	6	24,510	24.5
65–74	4	8786	45.5	0	9576	0	4	18,362	21.8
75–84	2	5142	38.9	1	6622	15.1	3	11,764	25.5
85+	2	1296	154.3	1	2369	42.2	3	3665	81.9
Total	12	144,287	8.3	14	144,184	9.7	26	288,471	9

*Prevalence expressed per 100,000 inhabitants per year.

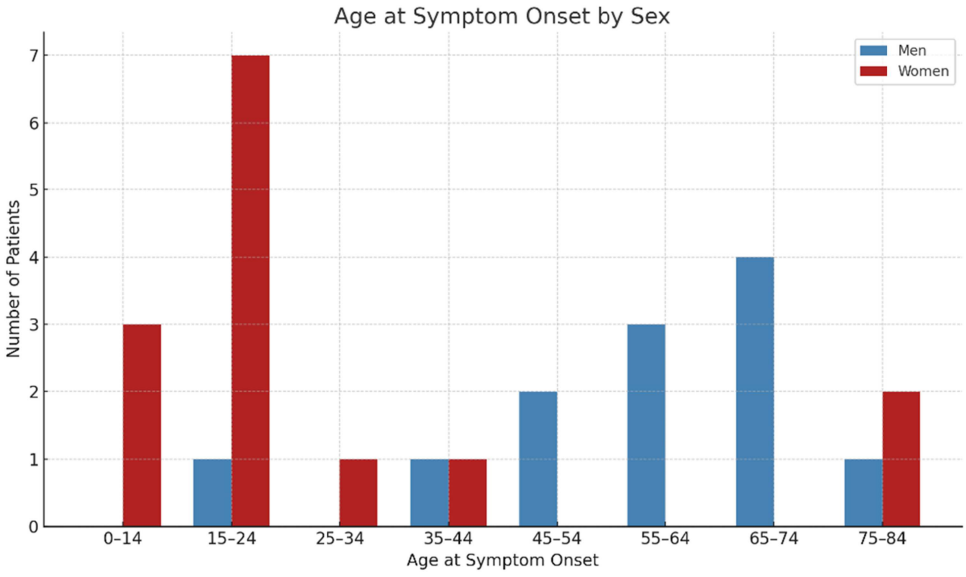


Figure 1. Age distribution of patients ($n = 26$) at symptom onset.

Incidence

During the five-year period from 31 December 1997, to 31 December 2002, a total of 11 patients were diagnosed with MG, corresponding to an annual incidence rate of 0.78 per 100,000 inhabitants.

Temporal gap between symptom onset and diagnosis

The median diagnostic delay was approximately 8 months. Eleven patients (42%) were diagnosed within six months of symptom onset, and 18 (69%) within one year. In five patients (19%), the delay exceeded three years, with the longest delay being 10 years. The mean age at symptom onset was 43 years (range: 5–84, median: 42); for men, 59 years (range: 24–79, median: 60) and for women, 29 years (range: 5–84, median: 20). Symptoms began before the age of 45 in 12 of 14 women (86%) and after the age of 50 in 10 of 12 men (83%) (see Figure 1).

Diagnosis confirmation

In 22 patients the clinical diagnosis was supported by positive antibody testing with acetylcholine receptor antibodies (AChRAb). Eighteen had a positive response to acetylcholinesterase inhibitor testing (Tensilon), and 15 had characteristic findings on electrophysiological studies. Figure 2 illustrates the overlap between diagnostic modalities confirming the diagnosis. Numbers indicate patients with positive results in each

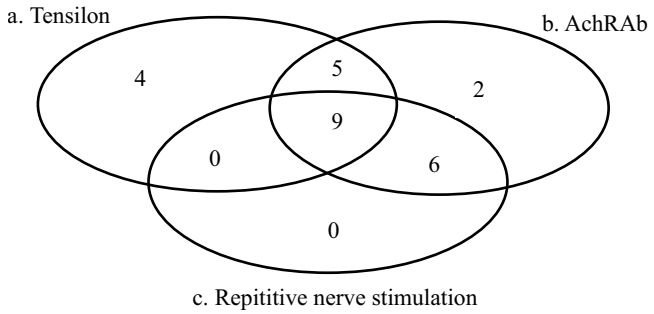


Figure 2. Overlap between diagnostic tests used in the evaluation of myasthenia gravis. The Venn diagram shows the distribution and overlap of positive findings for Tensilon testing, acetylcholine receptor antibody (AChRAb) testing and repetitive nerve stimulation among 26 patients.

diagnostic category. Nine patients were positive across all three modalities, while AChRAb testing identified the largest number of cases, including patients not detected by Tensilon testing or neurophysiological testing alone. Testing for muscle-specific kinase (MuSK) antibodies was not available in Iceland during the study period.

Age at diagnosis

The mean age at diagnosis for all patients was 58 years (range: 13–89 years, median: 61 years); for men, the mean was 70 years (range: 44–86 years, median: 73 years) and for women, 48 years (range: 13–89 years, median: 47 years). Two patients were younger than 15 years at the time of diagnosis.

The median age at diagnosis in the generalized MG group was 47.5 years (IQR: 27.5–60.8; range: 13–73), compared to 55.0 years (IQR: 45.5–64.5; range: 24–89) in the ocular MG group. While there was a trend toward younger age among patients with generalized MG, the interquartile ranges overlapped and a Mann–Whitney U test indicated no significant difference in the distribution of ages at diagnosis between groups ($U = 38.0$, $p = 0.22$). Figure 3 presents a combined violin and boxplot showing age at diagnosis by MG type.

Disease course

In 19 patients (73%), ocular symptoms were part of the initial disease presentation. Among 15 patients (58%) who initially only had ocular symptoms, 7 later developed generalized MG. Thirteen patients (50%) had experienced exacerbations with generalized symptoms at some point during their illness, including one case that needed mechanical ventilation. In total, 22 patients (85%) had been hospitalized due to MG, with 15 (58%) requiring hospitalization more than once. Table 2 shows the classification of patients as assessed in February 2003. Seven patients (27%) were asymptomatic and not receiving any treatment. In five patients (19%)—all men—symptoms were confined to the eyes. Fourteen patients (54%) had generalized symptoms; in five of these cases (19%), the symptoms were severe enough to interfere with work capacity and daily functioning.

Treatment in 2003

Twelve patients (46%) were not receiving any pharmacological therapy at the time of data collection. Acetylcholinesterase inhibitors were used regularly by 11 patients (42%); among them, 5 (19%) relied on this treatment alone, 4 (17%) combined it with azathioprine and 2 (8%) received regular intravenous immunoglobulin (IVIG) in addition. Based on contemporary definitions of refractory MG (persistent symptoms despite adequate immunosuppressive therapy or need for repeated rescue therapy), two patients (8%) required regular IVIG despite immunosuppressive therapy and could retrospectively be classified as having treatment-refractory MG according to contemporary definitions.

MG Type	Median Age, years	IQR (Q1–Q3)	Range (min–max, years)
Generalized MG	47.5	27.5 – 60.75	13 – 73
Ocular MG	55.0	45.5 – 64.5	24 – 89

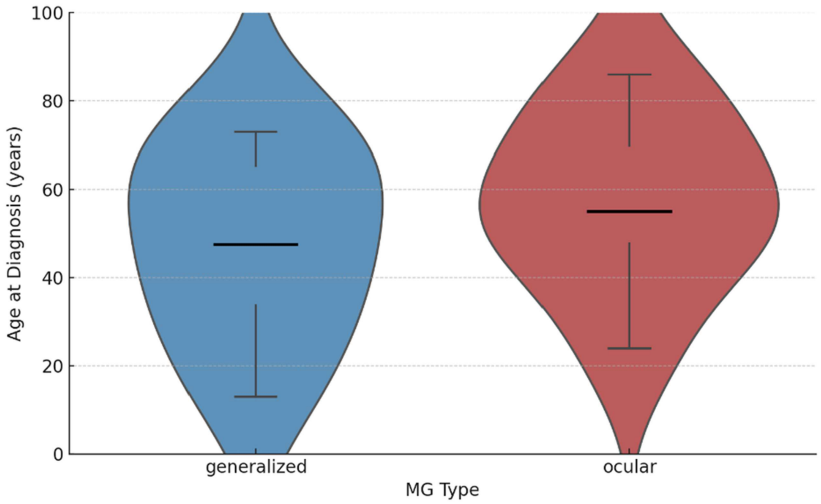


Figure 3. Age at Diagnosis by MG Type. The Seaborn violin plot illustrates the age distribution for generalized and ocular MG in Iceland, a small Arctic population, combining kernel density estimates with overlaid boxplots that display medians and interquartile ranges. Whiskers indicate the observed minimum and maximum ages in each group, with the y-axis limited to 0–100 years for clarity.

Table 2. Distribution of patients in February 2003 by Osserman classification and retrospective MGFA, by sex ($n = 26$).

Osserman classification	Retrospective MGFA class*	Men ($n = 12$)	%	Women ($n = 14$)	%	Total ($n = 26$)	%
A – asymptomatic (remission)	0	3	25	4	28.5	7	27
I – ocular	I	5	42	0	0	5	19
IIA – mild generalized	II–III (approx.)	3	25	6	43	9	35
IIB – moderate generalized	III–IV (approx.)	1	8	4	28.5	5	19
III – acute fulminating	IV–V (approx.)	0	0	0	0	0	0
IV – severe generalized	V	0	0	0	0	0	0
Total		12	100	14	100	26	100

Note: The table shows sex distribution by Osserman stage at assessment (February 2003). Osserman A indicates asymptomatic status without MG-specific medication at assessment (complete remission in this study). Osserman I indicates ocular-only involvement; Osserman IIA and IIB indicate generalized disease of increasing severity; Osserman III–IV indicate severe generalized disease with bulbar and/or respiratory involvement. MGFA classes are defined as follows: Class 0; Post-intervention status: complete stable remission (CSR); Class I: Any ocular muscle weakness; all other muscle strength normal; Class II: Mild weakness affecting muscles other than ocular (may also have ocular weakness); Class III: Moderate weakness affecting muscles other than ocular; Class IV: Severe weakness affecting muscles other than ocular; Class V: Defined by intubation or need for mechanical ventilation (excluding routine postoperative ventilation). Because the original clinical records were classified using the Osserman system, MGFA categories represent an approximate crosswalk rather than a formal MGFA assessment, intended to facilitate comparison with contemporary literature.

Three patients (12%) were treated exclusively with corticosteroids. Overall, 18 patients (69%) had received immunosuppressive therapy at some point during the disease course, with corticosteroids being the most frequently used agent (15 patients; 58%).

Plasmapheresis was administered to 3 patients (12%) in response to severe disease exacerbation, including one who underwent repeated procedures. Thymectomy had been performed in 11 patients (42%). Histopathological examination of thymic tissue revealed lymphoid hyperplasia in 3 of the 11 cases (27%) and thymic tumour in one case (9%). Of the 12 patients not receiving pharmacological treatment (46%), 7 (27%) were asymptomatic and fulfilled criteria for complete remission. The remaining 5 patients had mild persistent symptoms, primarily ocular, but did not require ongoing pharmacological therapy at the time of evaluation.

Discussion

This population-based study provides comprehensive information on the epidemiology of MG in Iceland. Although unpublished until now, the data retain scientific value adding to knowledge of MG in the Arctic Circle and serves as a benchmark for future longitudinal analyzes. In rare disease research, particularly within small, genetically homogeneous populations, legacy datasets are important for tracking epidemiological trends and informing health system planning. Over the past two decades, MG prevalence has more than doubled in many regions, driven by improved diagnostics and demographic shifts [13,22]. The observed prevalence of MG in Iceland (9.0 per 100,000) represents a notable increase from earlier national estimates: 6.4 per 100,000 in 1963 [17] and 6.8 per 100,000 in 1991 [18]. This upward trend likely reflects a combination of improved survival, attributable to advances in treatment, as seen in other Nordic countries [23,24], alongside increased case detection. These speculations are supported by the rise in incidence from 0.41 per 100,000/year during 1954–1963 to our findings 0.78 per 100,000/year during 1997–2002 [17,18]. Importantly, antibody testing was not available during the earliest Icelandic epidemiological study in 1963. The introduction of acetylcholine receptor antibody assays in Iceland around the mid-1980s likely improved diagnostic sensitivity and case ascertainment, particularly among patients with milder or atypical presentations. This development may have contributed to the observed increase in both prevalence and incidence over time. Earlier case identification relied primarily on clinical examination and pharmacological testing, which may have led to underdiagnosis.

Demographic changes, particularly an aging population, may play a role, as late-onset MG has become more prevalent [22]. Since men are more frequently affected at older ages [10,25], this may partly explain the higher proportion of men in our cohort (male-to-female ratio: 1:1.2), which contrasts with previous Icelandic and international studies reporting ratios closer to 1:2 or 1:3 [26,27]. Overall, the prevalence figure in our study is consistent with those reported in many international studies (see Table 3) [11,23,24,26,28–32].

As shown in Table 3, three studies report notably higher MG prevalence rates (14–15 per 100,000). The elevated prevalence observed in the Virginia study may partly reflect the inclusion of a higher proportion of black individuals, among whom MG is thought to occur more frequently [31]. In the Cambridgeshire study, less stringent diagnostic criteria were applied compared to other studies, including our own, likely contributing to the higher figures [26]. In Stockholm, the existence of a specialized MG centre with dedicated personnel likely improved both diagnostic rates and patient follow-up [11]. However, the presence of such a referral centre may also have introduced bias, as patients from other parts of the country could have relocated to Stockholm for specialized care, potentially inflating the reported regional prevalence. This issue was not specifically addressed in the Swedish study. Nevertheless, the high prevalence reported raises the possibility that our own study may not have captured all diagnosed MG cases in Iceland.

Our results can also be compared with a recent systematic review and meta-analysis which estimated the global prevalence of MG to be 12.4 per 100,000 (95% CI: 10.6–14.5), with slightly lower figures reported in continental Europe (approximately 10 per 100,000) [33]. Other contemporary studies report rising rates of MG worldwide [34,35]. The precise drivers of these increased trends, whether improved diagnostic sensitivity, heightened clinical awareness, or demographic shifts, remain an area of active and future

Table 3. Incidence and prevalence of myasthenia gravis according to previous epidemiological studies.

Region	Year	Incidence*	Prevalence*
Finland [28]	1974	–	5.2
Kumamoto, Japan [29]	1982	–	6.7
The Netherlands [32]	1992	0.46	7.3
Denmark [24]	1988	0.44	7.7
Norway [23]	1981	0.4	9
Sardinia, Italy [30]	1994	0.25	11.1
Stockholm, Sweden [11]	1998	–	14.1
Virginia, United States [31]	1984	0.91	14.2
Cambridgeshire, United Kingdom [26]	1997	1.1	14.6

* Incidence expressed per 100,000 inhabitants per year.

Note: For ease of comparison, studies are presented in order of increasing reported prevalence, rather than by author name or publication year.

investigations. In this context, our study offers a meaningful historical base comparing a rare disease in a small Arctic area which aligns with broader global trends.

Patients presenting with clear ocular or bulbar symptoms were generally diagnosed without significant delay. In contrast, patients with more nonspecific limb or facial symptoms experienced longer diagnostic delays. Notably, one patient was misattributed with "chronic fatigue" for five years before the emergence of ocular signs prompted reevaluation, while another was treated for anxiety for three years prior to the eventual diagnosis of MG.

Although 13 patients experienced disease exacerbations during the course of illness, the overall prognosis in this cohort was relatively favourable. At the time of data collection, 46% of patients were either asymptomatic or had symptoms mild enough to require no ongoing medication, a notably higher proportion than reported in the Swedish (12%) [11,35] and British (16%) cohorts [26]. This comparatively positive outcome may reflect early recognition of typical ocular symptoms, effective long-term disease management and the availability of specialist care within Iceland's centralized healthcare system. Overall, treatment strategies in Iceland were broadly aligned with international standards, although steroid therapy appeared to be used somewhat more frequently. The proportion of patients undergoing thymectomy (42%) was comparable to figures reported elsewhere, including 57% in Sweden and 35% in the United Kingdom [11,26], suggesting that local clinical practice was consistent with contemporaneous European trends.

The high proportion of patients requiring hospitalization (85%) may appear discordant with the generally favourable long-term outcomes observed. However, during the study period, hospital admission was frequently required for diagnostic confirmation (e.g. edrophonium testing), initiation of immunosuppressive therapy, and monitoring during exacerbations. In a small healthcare system with centralized specialist services, inpatient evaluation was often standard practice rather than an indicator of severe disease.

At the time of the study, edrophonium (Tensilon) testing remained in clinical use in Iceland. Although this pharmacologic challenge test has since been discontinued due to the risk of rare but serious adverse effects, no such complications were observed in this cohort, and its diagnostic utility was evident, contributing to confirmation in 18 patients. Since then, diagnostic practices have evolved substantially. Advances in neurophysiological methods, including broader access to single-fibre electromyography (EMG) and standardized training in repetitive nerve stimulation, have markedly improved the accuracy and reliability of MG diagnosis in Iceland.

Strengths and limitations

Historical epidemiological data are often prone to bias due to population heterogeneity and geographic variation [36]. The small cohort in our study limits the strength of the data but at the same time Iceland offers an unusually robust context for such research, with a genetically and socially homogeneous population and comprehensive national health records. This infrastructure allows for near-complete case ascertainment across both primary and specialist care. Although the response rate from general practitioners was limited, some using ICD-code searches, others relying on recall, all reported cases were also documented in medical records and confirmed by neurologists. This high concordance, combined with direct contact with all identified MG patients through in-person or telephone interviews, strengthens the reliability of the data and the validity of the prevalence estimates. Additionally, MGFA classification was retrospectively approximated from Osserman staging, which may introduce minor misclassification compared with contemporaneous MGFA assessment.

Conclusion

This study provides valuable epidemiological and clinical insights into MG within the Icelandic population and contributes to the enduring importance of historical datasets in rare disease research. Such data plays a role in strengthening national surveillance systems, enabling longitudinal comparisons and informing global understanding of disease patterns, considering small and geographically distinct populations such as those in the Arctic region.

Author contributions

CRediT: **Haukur Hjaltason:** Conceptualization, Formal analysis, Methodology, Writing – original draft, Writing – review & editing; **Haraldur Ólafsson:** Conceptualization, Data curation, Writing – review & editing; **Finnbogi Jakobsson:** Methodology, Writing – review & editing; **Marianne Elisabeth Klinke:** Formal analysis, Writing – original draft, Writing – review & editing; **Ólöf Elíasdóttir**

Disclosure statement

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Key statement

This nationwide legacy study assessed the epidemiology of myasthenia gravis in Iceland, revealing stable yet increasing prevalence and incidence, and demonstrating how centralized Arctic healthcare systems enable reliable rare disease surveillance in small, genetically homogeneous populations.

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